

A dark, atmospheric photograph of a person's silhouette standing on a rocky outcrop, looking out over a vast mountain range under a hazy, low-sun sky. The person is positioned in the center, slightly to the left, with their back to the camera. The mountains in the background are layered and shrouded in mist. The overall color palette is dark, with muted browns, greys, and a hint of orange from the low sun.

CUSTOMER SEGREGATION IN AIRLINES

Using Python

What Is Frequent Flyer Program?

- The Frequent Flyer Program is an incentive program operated by an airline to reward customers for their continued loyalty.
- As a traveler, you earn free miles for the miles that you fly on a particular airline.
- The concept behind Frequent Flyer Programs is that the airlines want their passengers to become lifetime customers.
- It is much more costly for the airlines to get new customers than it is to retain the ones they already have.
- The more frequently you fly with them, the greater your awards.

Provided Data Description

The following is the description of NZ airline's frequent flyer program dataset with 3999 entries and 8 variables.

Variables (in order as in dataset)	Description
FFP#	Frequent flyer program id
AwardMiles	Balance number of miles eligible for award travel
EliteMiles	Number of miles counted as qualifying for elite status
PartnerMiles	Number of miles earned from transactions while using partner's services in the past 12 months
PartnerTrans	Number of transactions made at partner's in the past 12 months
FlyingReturnsMiles	Number of flight miles in the past 12 months
FlightTrans	Number of flight transactions in the past 12 months
EnrollDuration	Number of days since enrolled for FFP

Problem Statement

- NZ Airlines has a frequent flyer program.
- NZA is in possession of a lot of data about the behavior of their FFP customers.
- NZA wishes to develop meaningful customer segments using the FFP behavior data.
- NZA will use the profile and size of customer segments to develop appropriate marketing and promotional programs to improve their revenues, and satisfy their customers better.

Data Pre-processing

- Obtaining inference using info and describe of the data, we get to know that all the variables are in numeric.
- It is seen that the dataset has no null values.
- Expect 'Awardmiles', 'EliteMiles', 'PartnerMiles' and 'FlyingReturnsMiles' other variables are normally distributed.
- the dataset does not contain duplicates.

```
data.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3999 entries, 0 to 3998
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   FFP#                   3999 non-null   int64
1   AwardMiles             3999 non-null   int64
2   EliteMiles             3999 non-null   int64
3   PartnerMiles           3999 non-null   int64
4   PartnerTrans           3999 non-null   int64
5   FlyingReturnsMiles     3999 non-null   int64
6   FlightTrans            3999 non-null   int64
7   EnrollDuration         3999 non-null   int64
dtypes: int64(8)
memory usage: 250.1 KB
```

```
data.describe()
```

	FFP#	AwardMiles	EliteMiles	PartnerMiles	PartnerTrans	FlyingReturnsMiles	FlightTrans	EnrollDuration
count	3999.000000	3.999000e+03	3999.000000	3999.000000	3999.000000	3999.000000	3999.000000	3999.000000
mean	2014.819455	7.360133e+04	144.114529	17144.846212	11.60190	460.055764	1.373593	4118.55939
std	1160.764358	1.007757e+05	773.663804	24150.967826	9.60381	1400.209171	3.793172	2065.13454
min	1.000000	0.000000e+00	0.000000	0.000000	0.000000	0.000000	0.000000	2.000000
25%	1010.500000	1.852750e+04	0.000000	1250.000000	3.000000	0.000000	0.000000	2330.000000
50%	2016.000000	4.309700e+04	0.000000	7171.000000	12.000000	0.000000	0.000000	4096.000000
75%	3020.500000	9.240400e+04	0.000000	23800.500000	17.000000	311.000000	1.000000	5790.500000
max	4021.000000	1.704838e+06	11148.000000	263685.000000	86.000000	30817.000000	53.000000	8296.000000

```
# checking for duplicates
len(data[data.duplicated()])

0
```

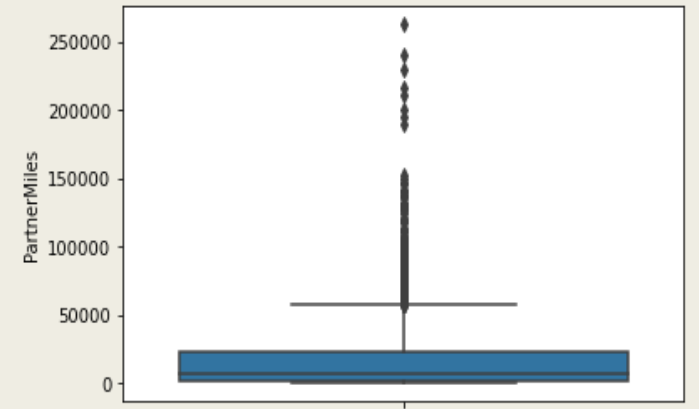
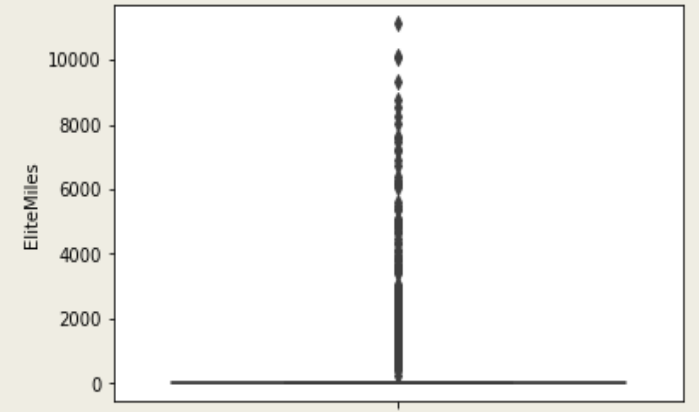
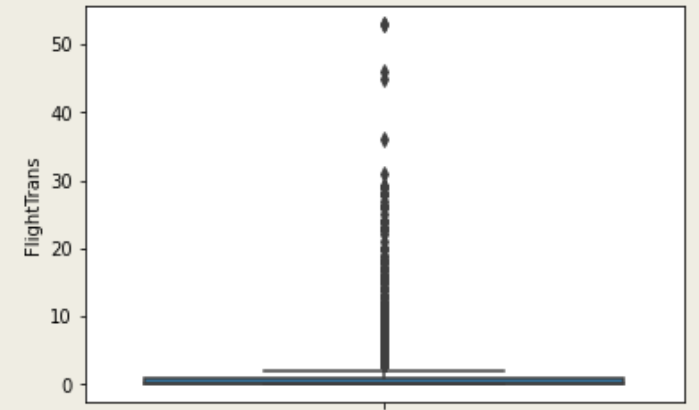
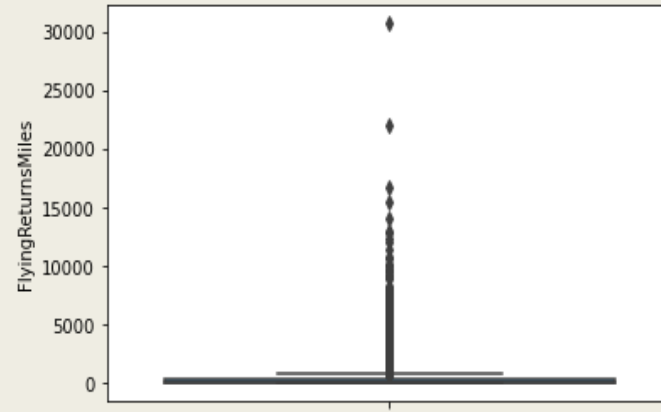
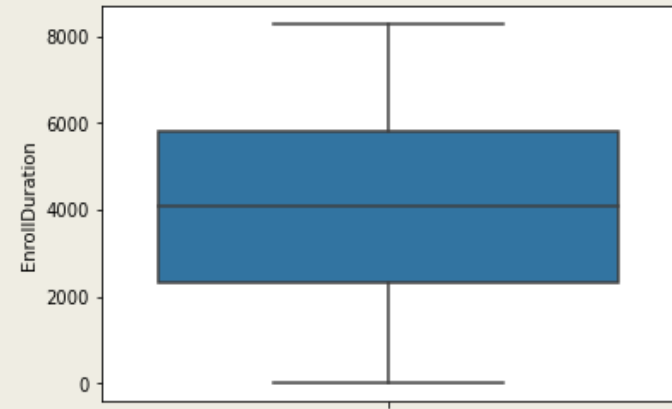
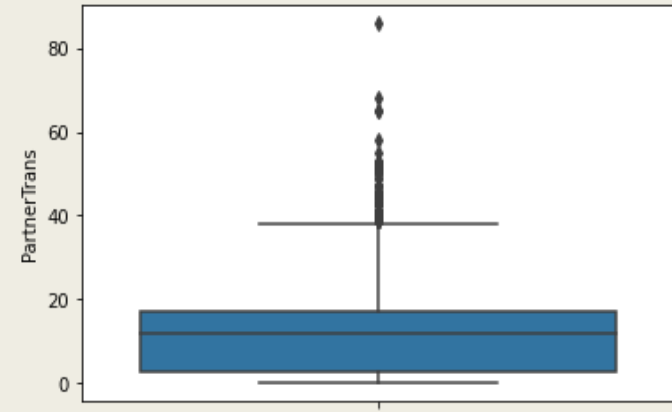
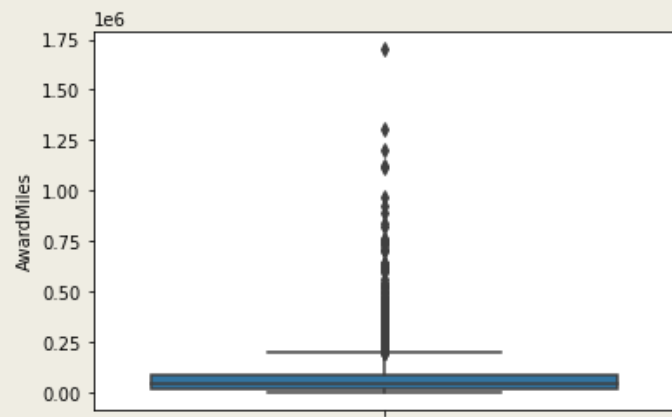
Data Pre-processing

- Removing the first column 'FFP#' as its just a id for the customers who are the part of Frequent Flying Program and it does not making any impact in the analysis.
- Changing the 'Awardmiles' values from exponential to proper numeric value.

	AwardMiles	EliteMiles	PartnerMiles	PartnerTrans	FlyingReturnsMiles	FlightTrans	EnrollDuration
count	3999.000	3999.000	3999.000	3999.000	3999.000	3999.000	3999.000
mean	73601.328	144.115	17144.846	11.602	460.056	1.374	4118.559
std	100775.665	773.664	24150.968	9.604	1400.209	3.793	2065.135
min	0.000	0.000	0.000	0.000	0.000	0.000	2.000
25%	18527.500	0.000	1250.000	3.000	0.000	0.000	2330.000
50%	43097.000	0.000	7171.000	12.000	0.000	0.000	4096.000
75%	92404.000	0.000	23800.500	17.000	311.000	1.000	5790.500
max	1704838.000	11148.000	263685.000	86.000	30817.000	53.000	8296.000

Viewing The Distribution Of The Variables Using The Boxplot:

- Boxplot helps us to visualize the outliers in a dataset because it show the median of a dataset along with the first and third quartiles.
- It also shows the skewness of the variable.
- Expect 'EnrollDuration' , all other variables have outliers.

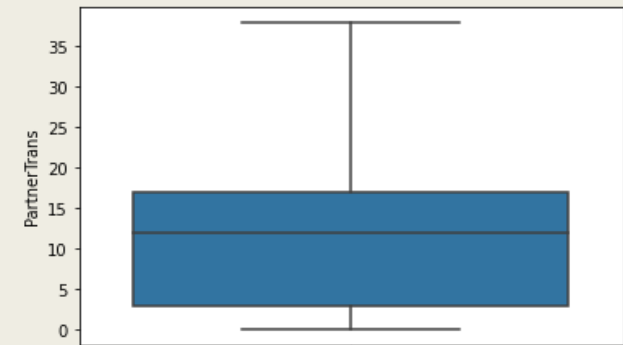
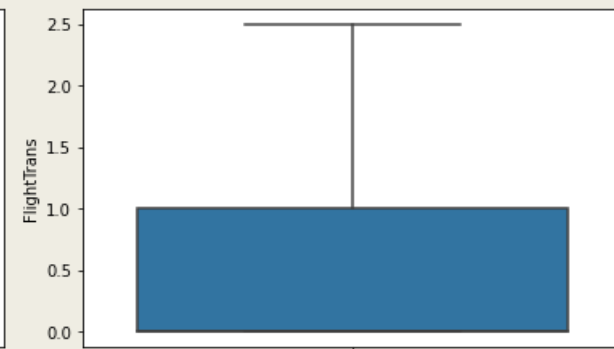
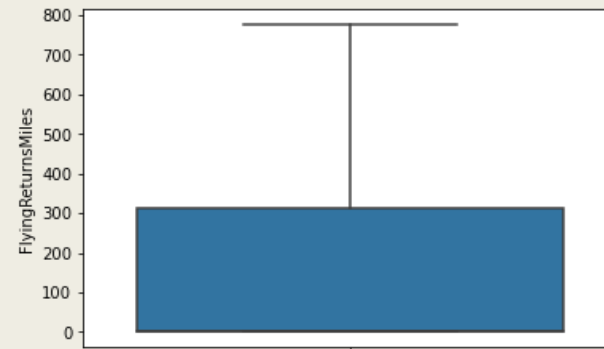
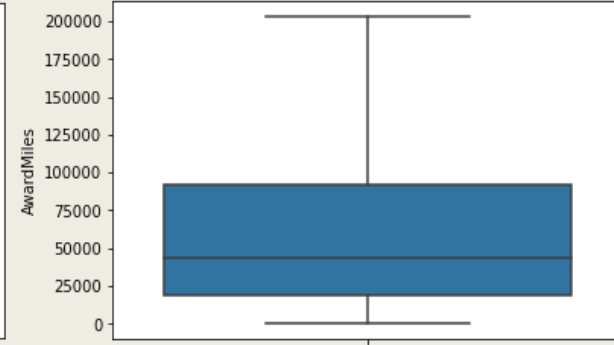
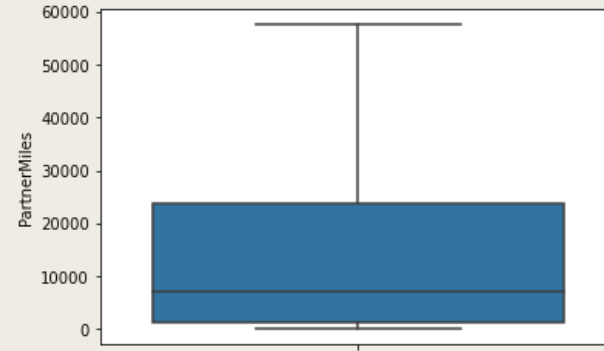


Data Pre-processing

Removing Outliers:

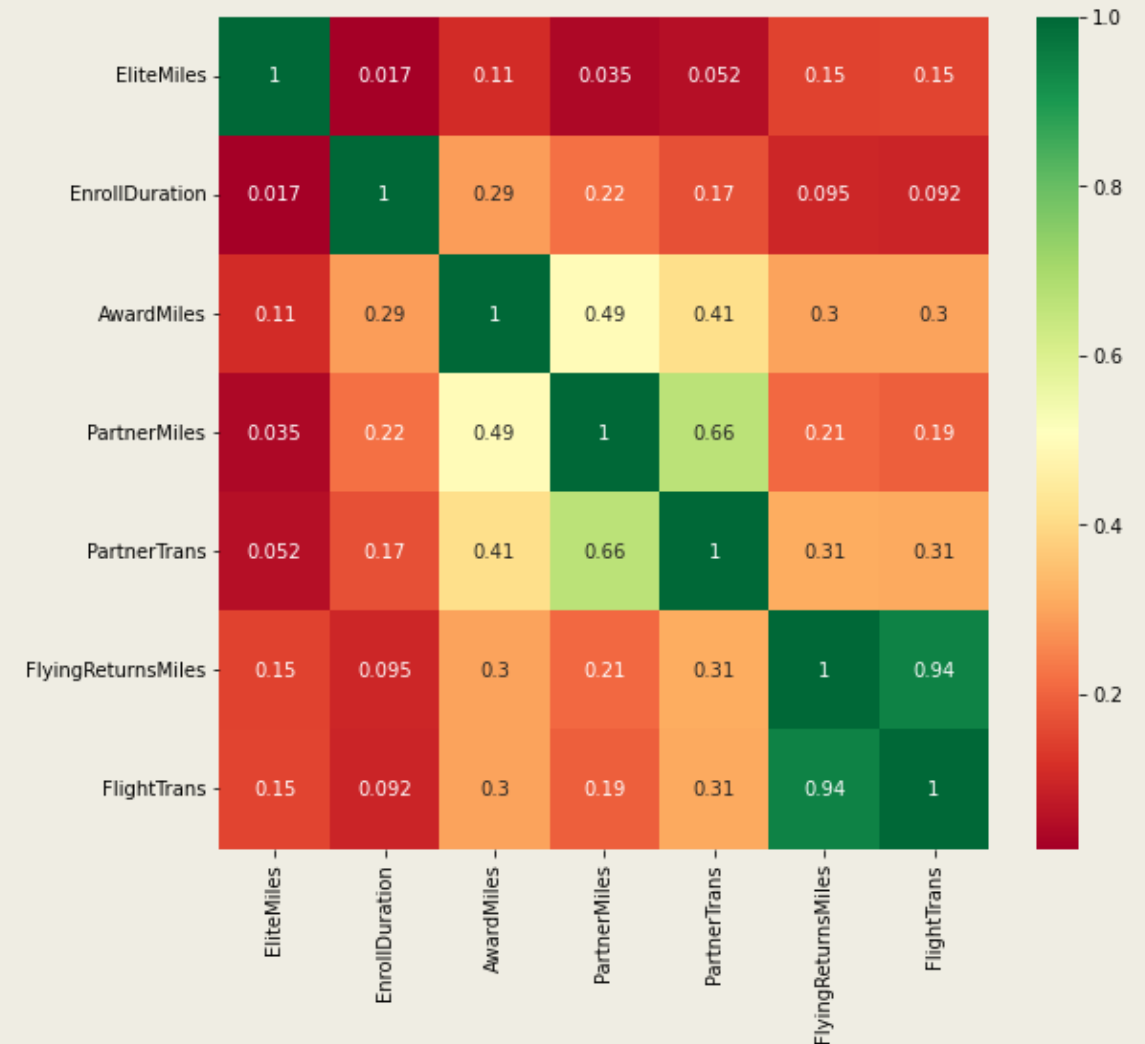
We are removing the outliers using median as the mean is affected by outliers.

We are not removing the outliers in the variable 'EliteMiles' as it could lead to loss of information when removing higher number of entries.



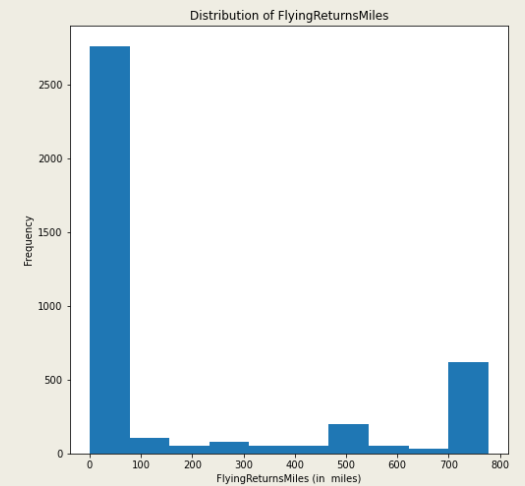
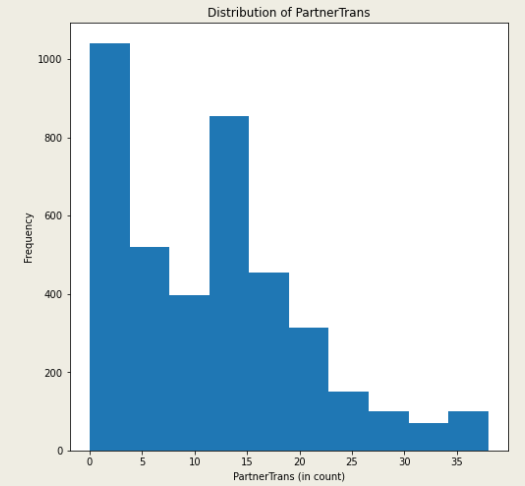
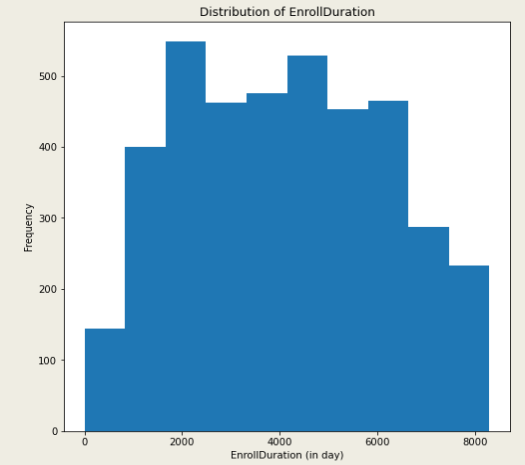
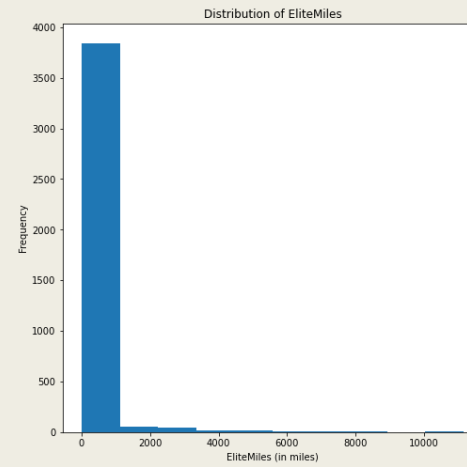
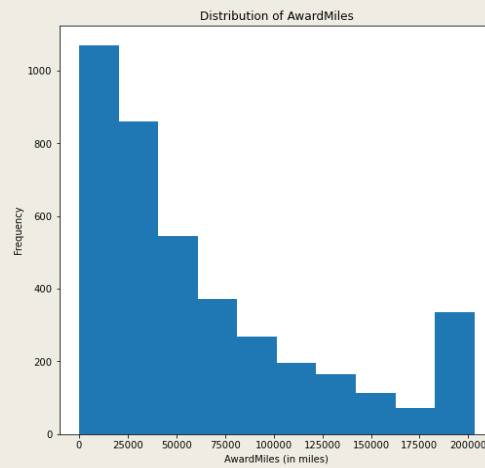
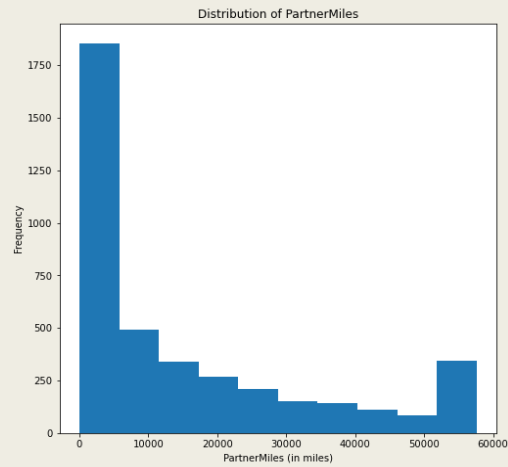
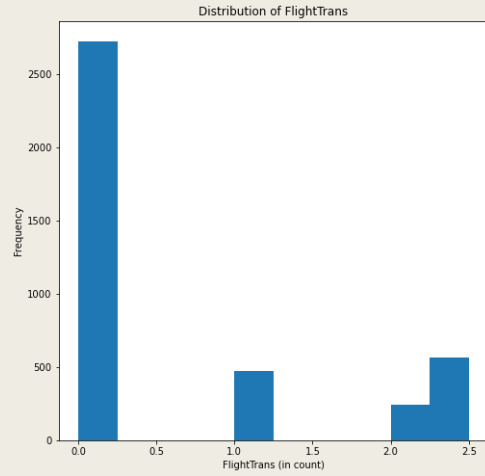
Explanatory Data Analysis

- FlyingReturnsMiles and FlightTrans are highly positively correlated. This implies as the FlightTrans increases FlightReturnsmiles.
- PartnerTrans and PartnerMiles are second highly positively correlated.
- AwardMiles is moderately correlated with PartnerMiles and PartnerTrans.
- FlightTrans and FlightReturnsMiles are moderately correlated with AwardMiles and PartnerTrans.
- EnrollDuration is not properly correlated with any of the variables in the data.



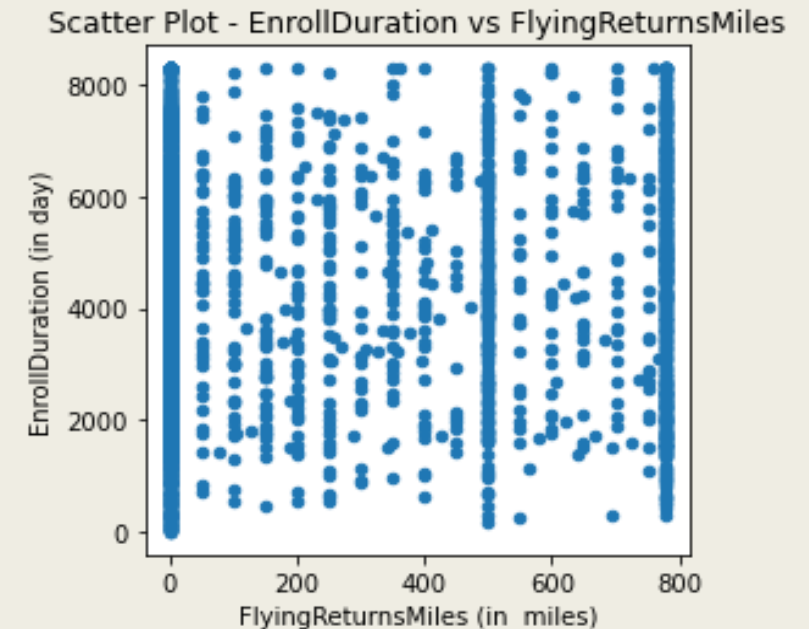
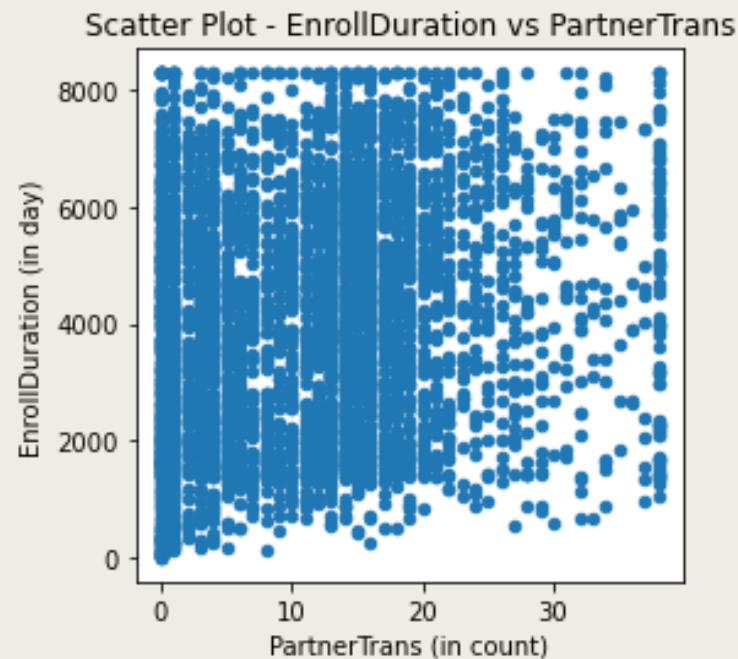
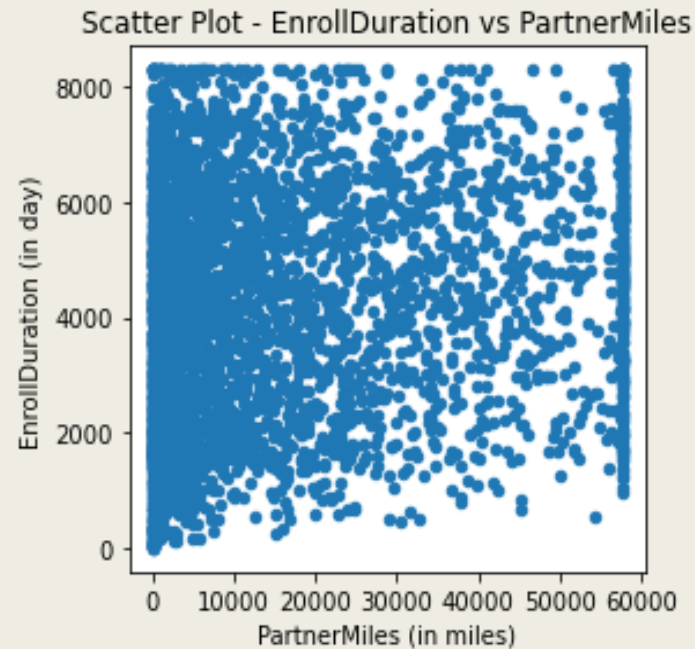
Explanatory Data Analysis

- Large Amount of people stayed in the program after enrolment between 2000 and 6000 days.
- Only few people travelled beyond 1000 miles and they will be considered in the process of qualifying to Elite Status.
- People transacted highly between 0 and 15 using Partner' Service.
- Most of the people travelled between 0 and 100 miles and better number of people travelled between 700 and 800 miles while flying back using same company's flight.
- Large number of people earned 0 to 5000 miles and only few people earned 40,000 to 57,000 miles while doing transactions using Partner's Service.
- Maximum number of people travelled only 0 to 50,000 miles to attain Award travel.
- Large number of flight transaction happen between 0.0 and 0.25.

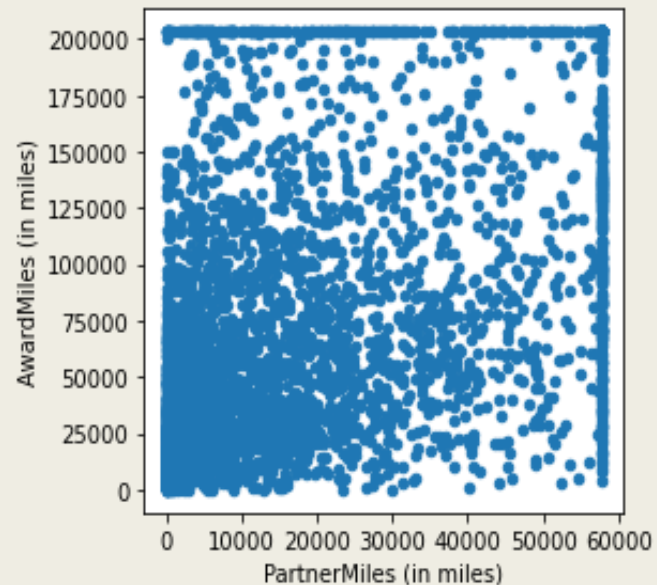


Explanatory Data Analysis

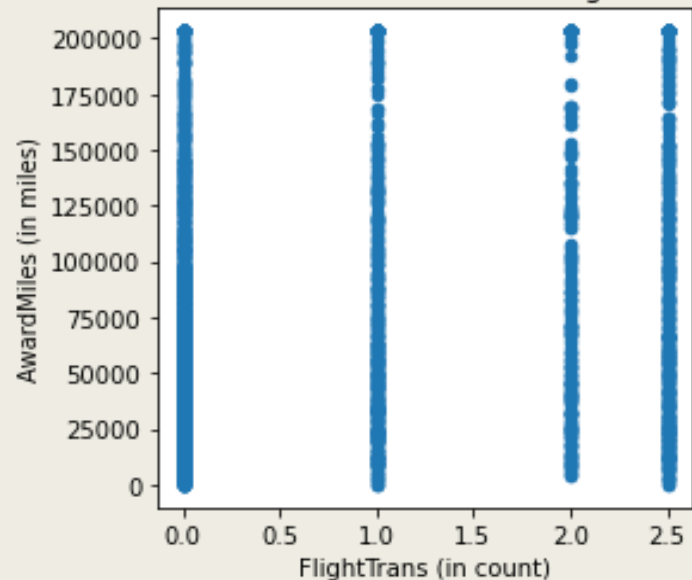
As the comparison plots between two different features of the data ended up with even distribution. So we could not get proper insights using those plots.



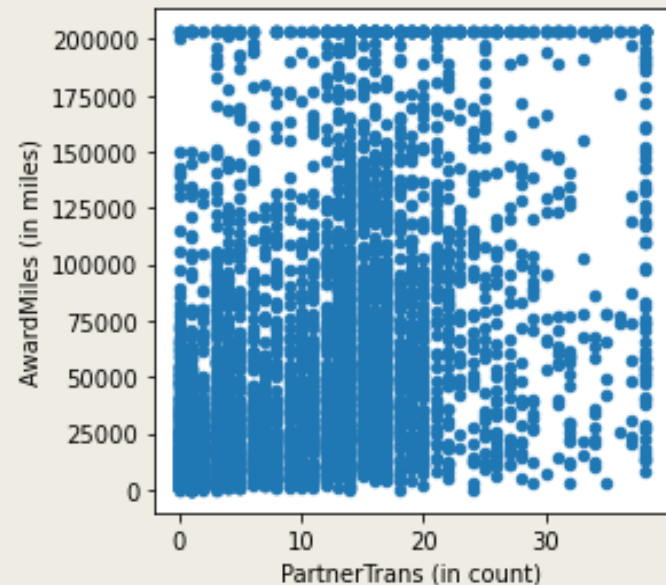
Scatter Plot - AwardMiles vs PartnerMiles



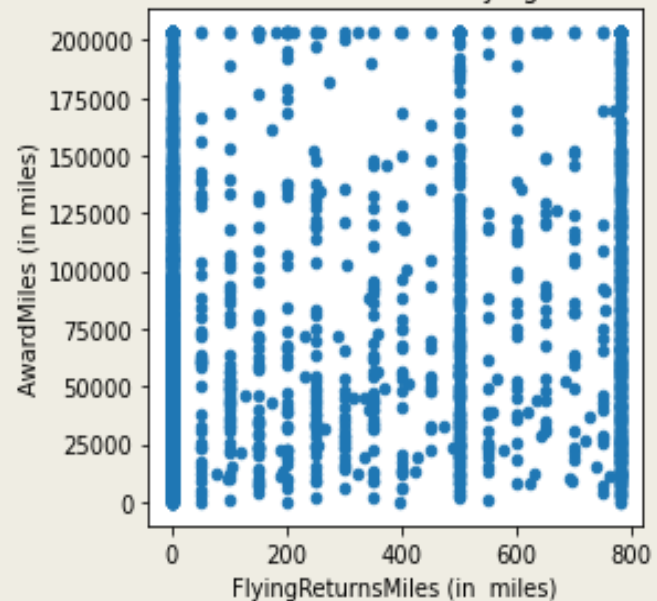
Scatter Plot - AwardMiles vs FlightTrans



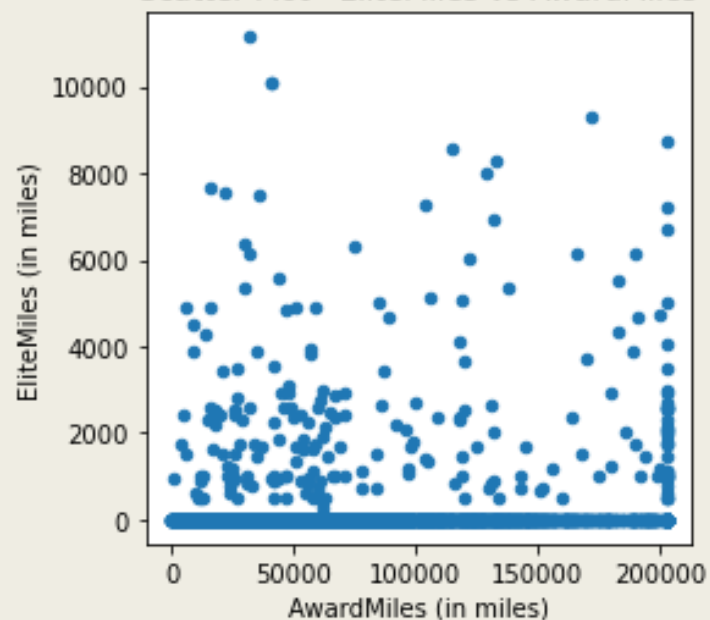
Scatter Plot - AwardMiles vs PartnerTrans



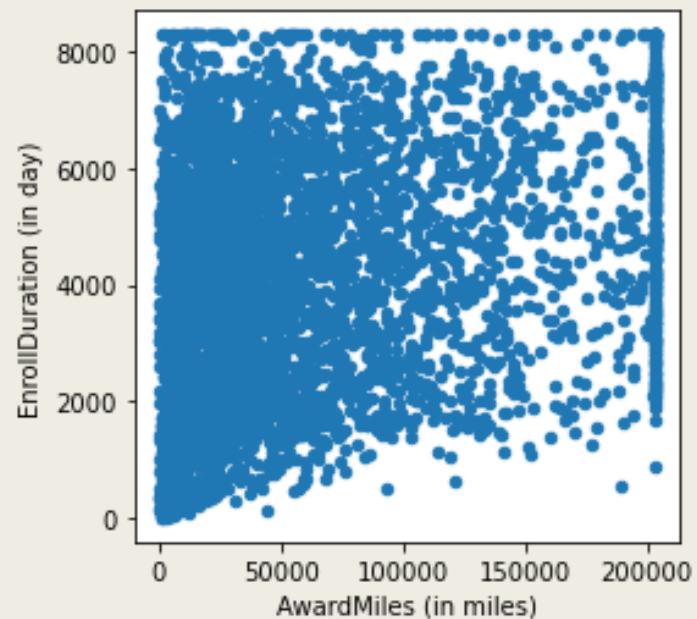
Scatter Plot - AwardMiles vs FlyingReturnsMiles



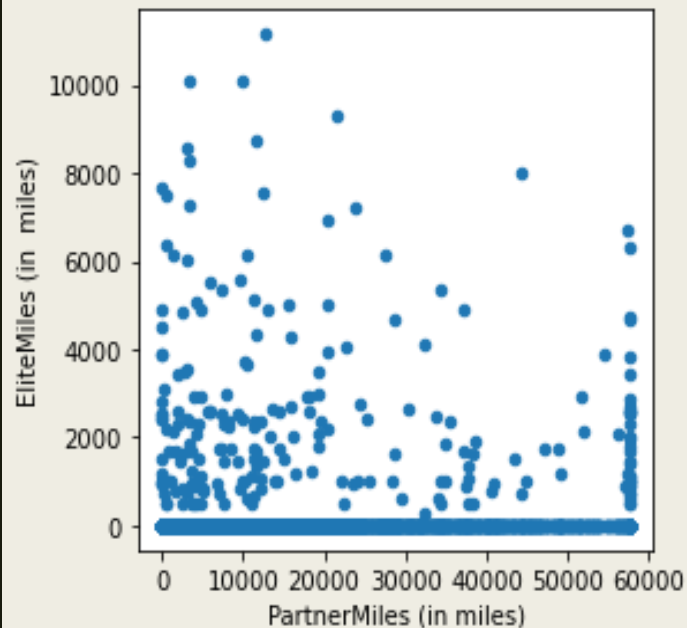
Scatter Plot - EliteMiles vs AwardMiles



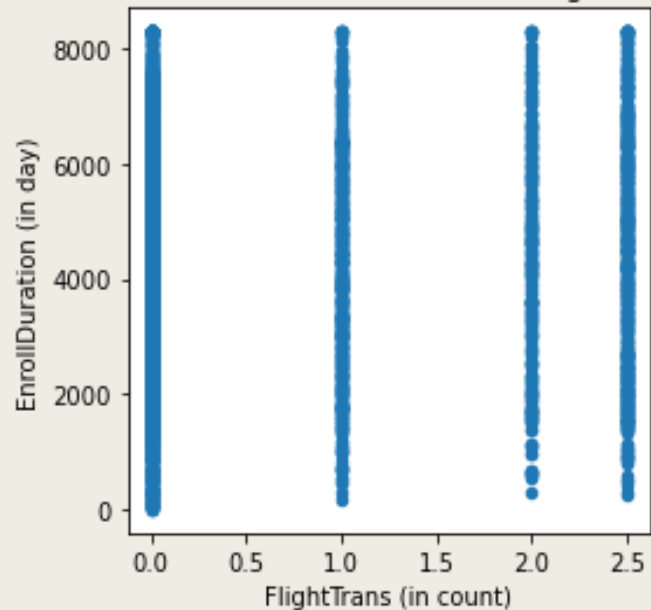
Scatter Plot - EnrollDuration vs AwardMiles



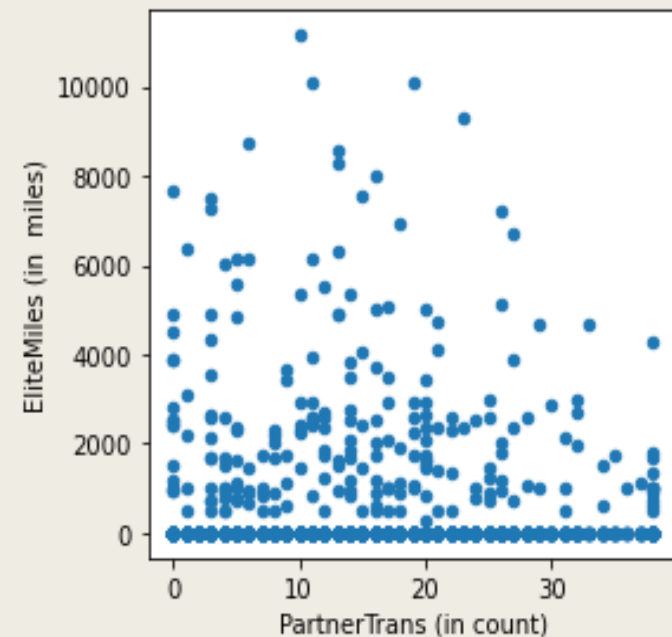
Scatter Plot - EliteMiles vs PartnerMiles



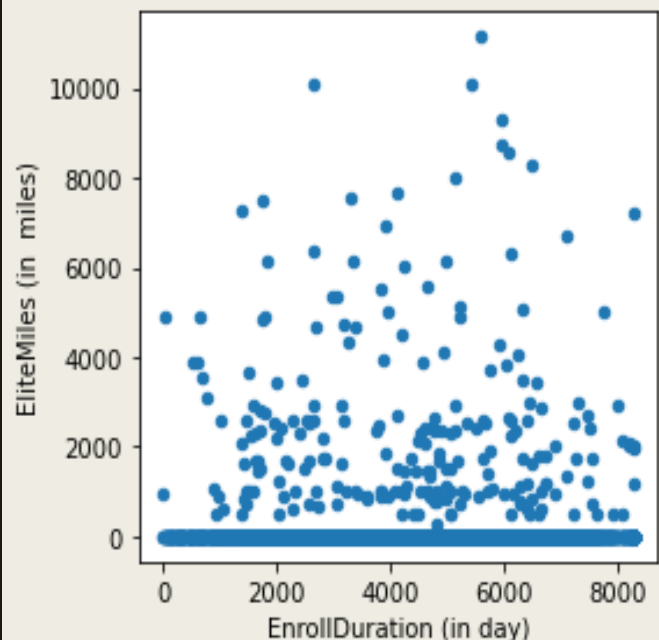
Scatter Plot - EnrollDuration vs FlightTrans



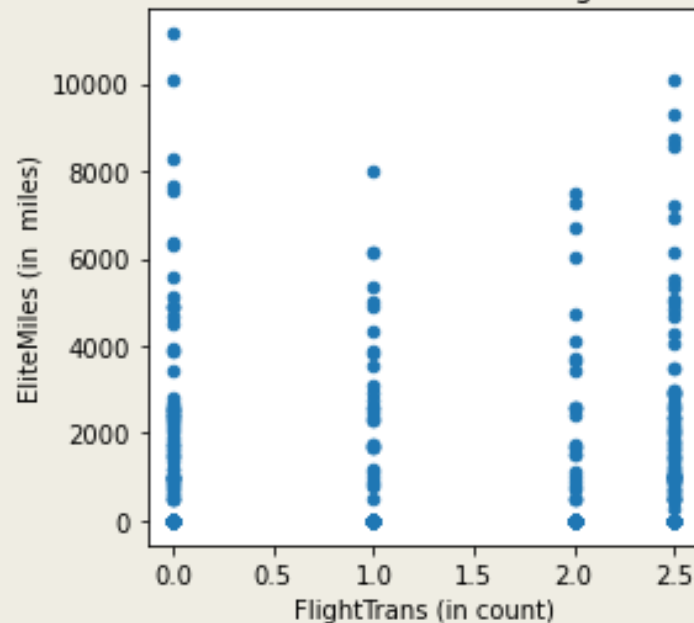
Scatter Plot - EliteMiles vs PartnerTrans



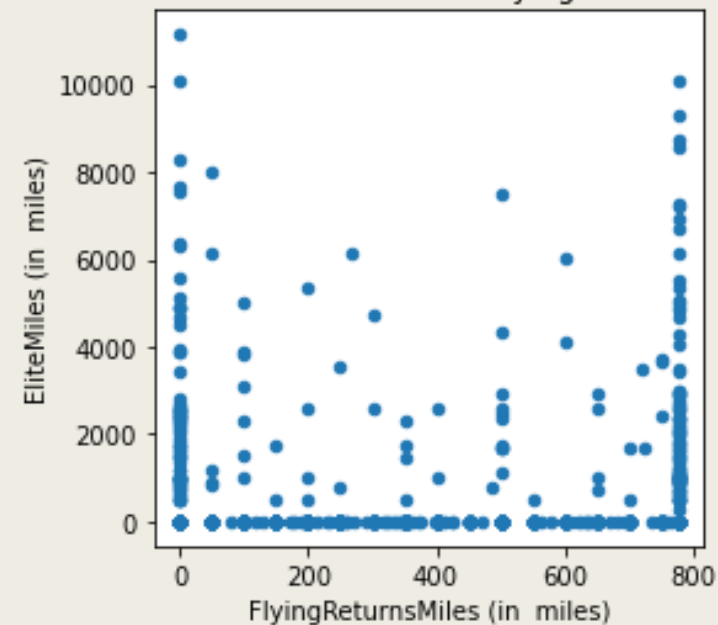
Scatter Plot - EliteMiles vs EnrollDuration



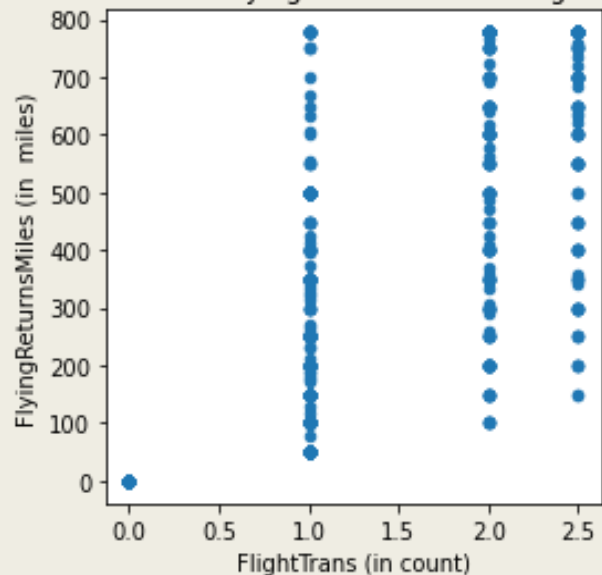
Scatter Plot - EliteMiles vs FlightTrans



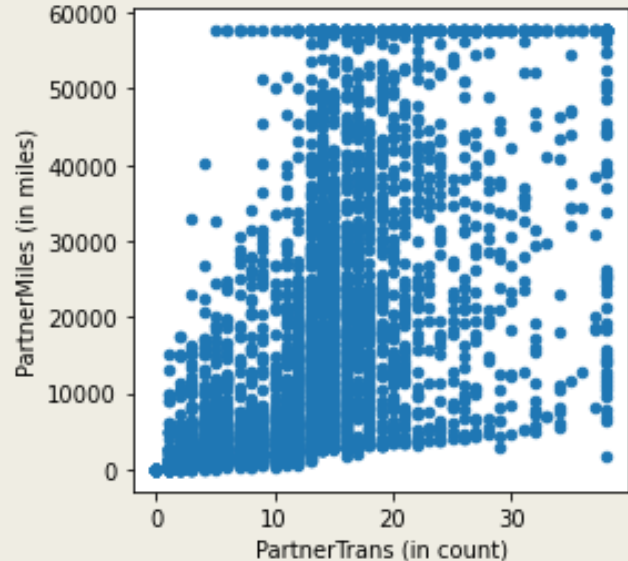
Scatter Plot - EliteMiles vs FlyingReturnsMiles



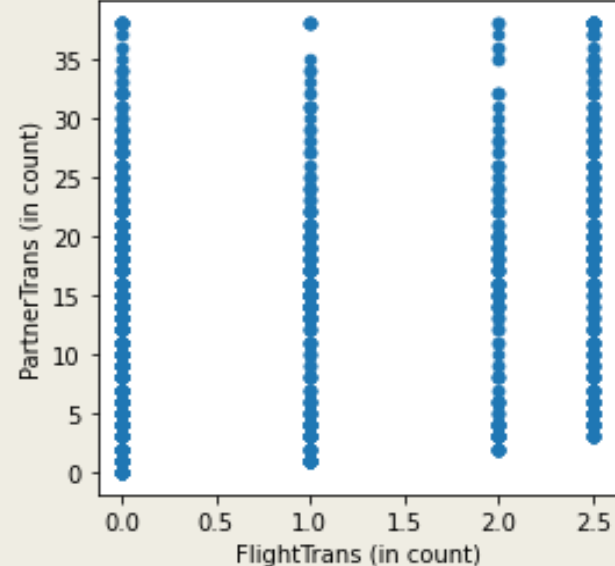
Scatter Plot - FlyingReturnsMiles vs FlightTrans



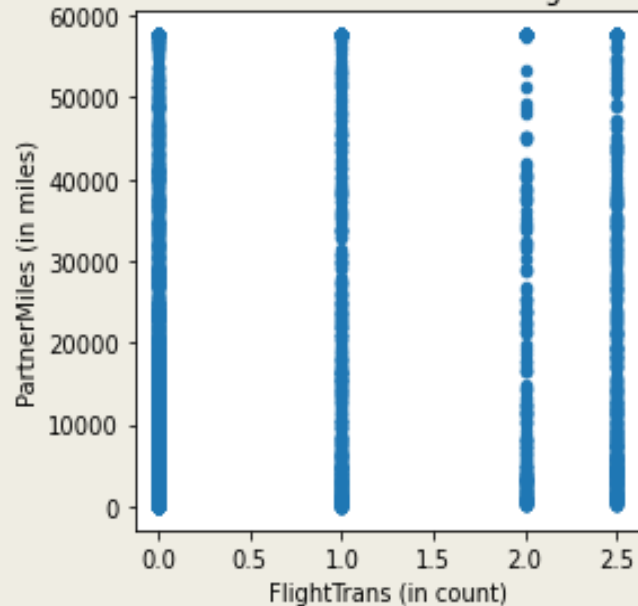
Scatter Plot - PartnerMiles vs PartnerTrans



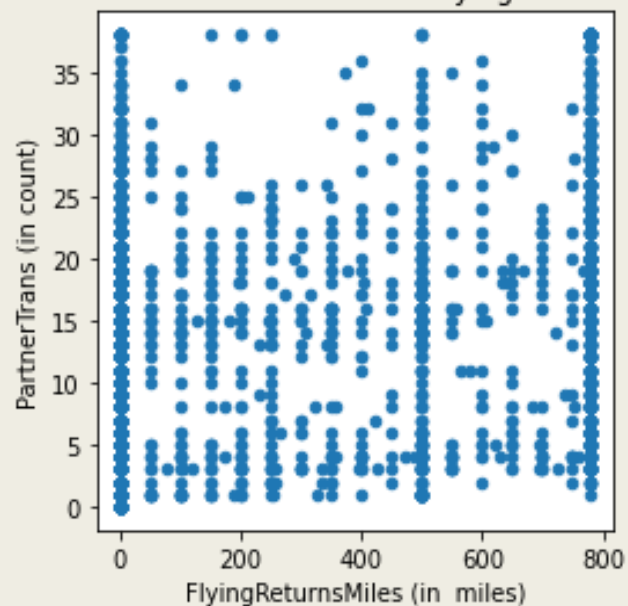
Scatter Plot - PartnerTrans vs FlightTrans



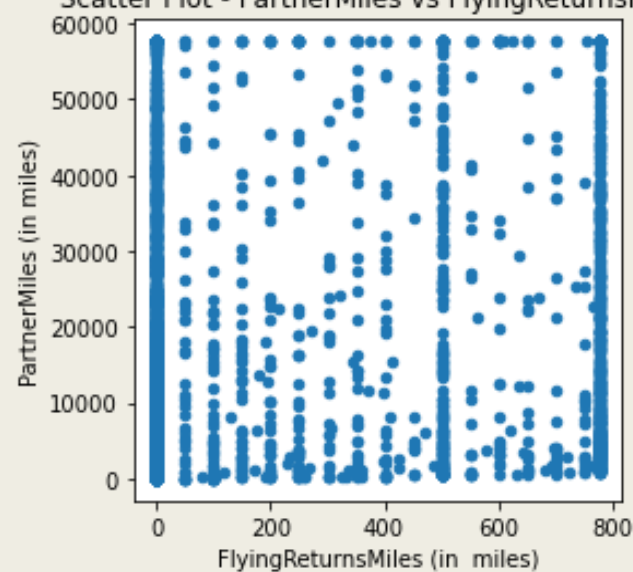
Scatter Plot - PartnerMiles vs FlightTrans



Scatter Plot - PartnerTrans vs FlyingReturnsMiles

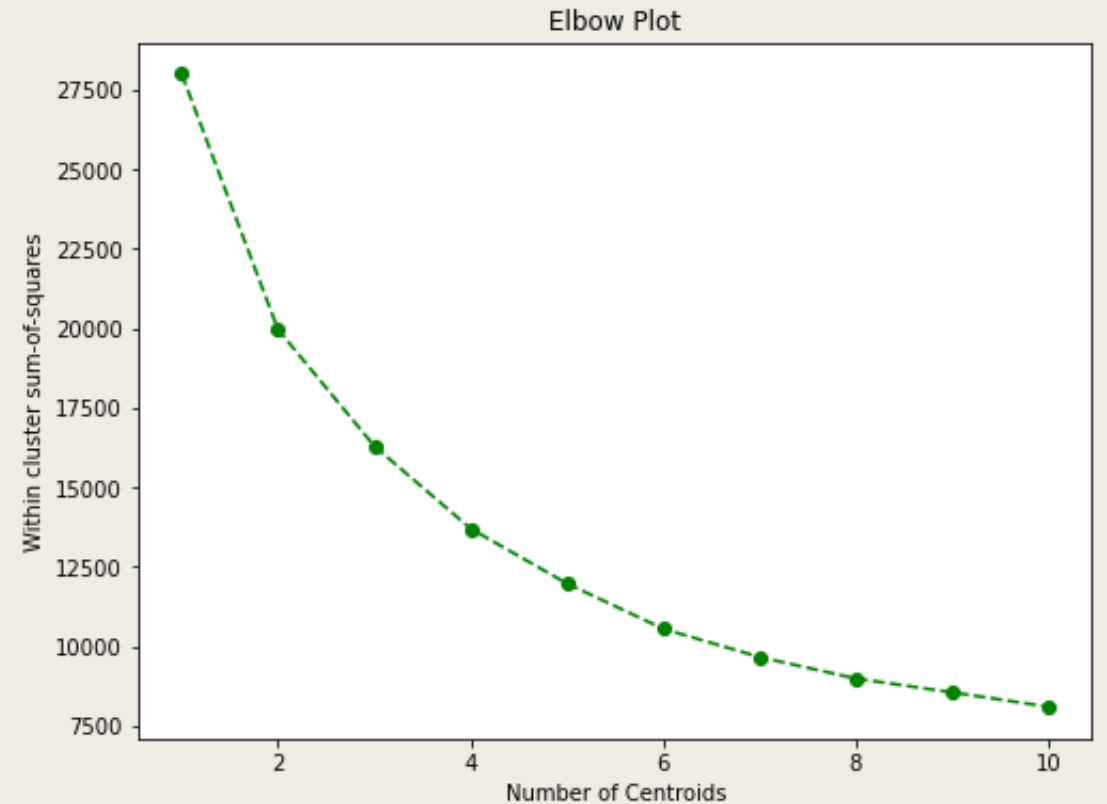


Scatter Plot - PartnerMiles vs FlyingReturnsMiles



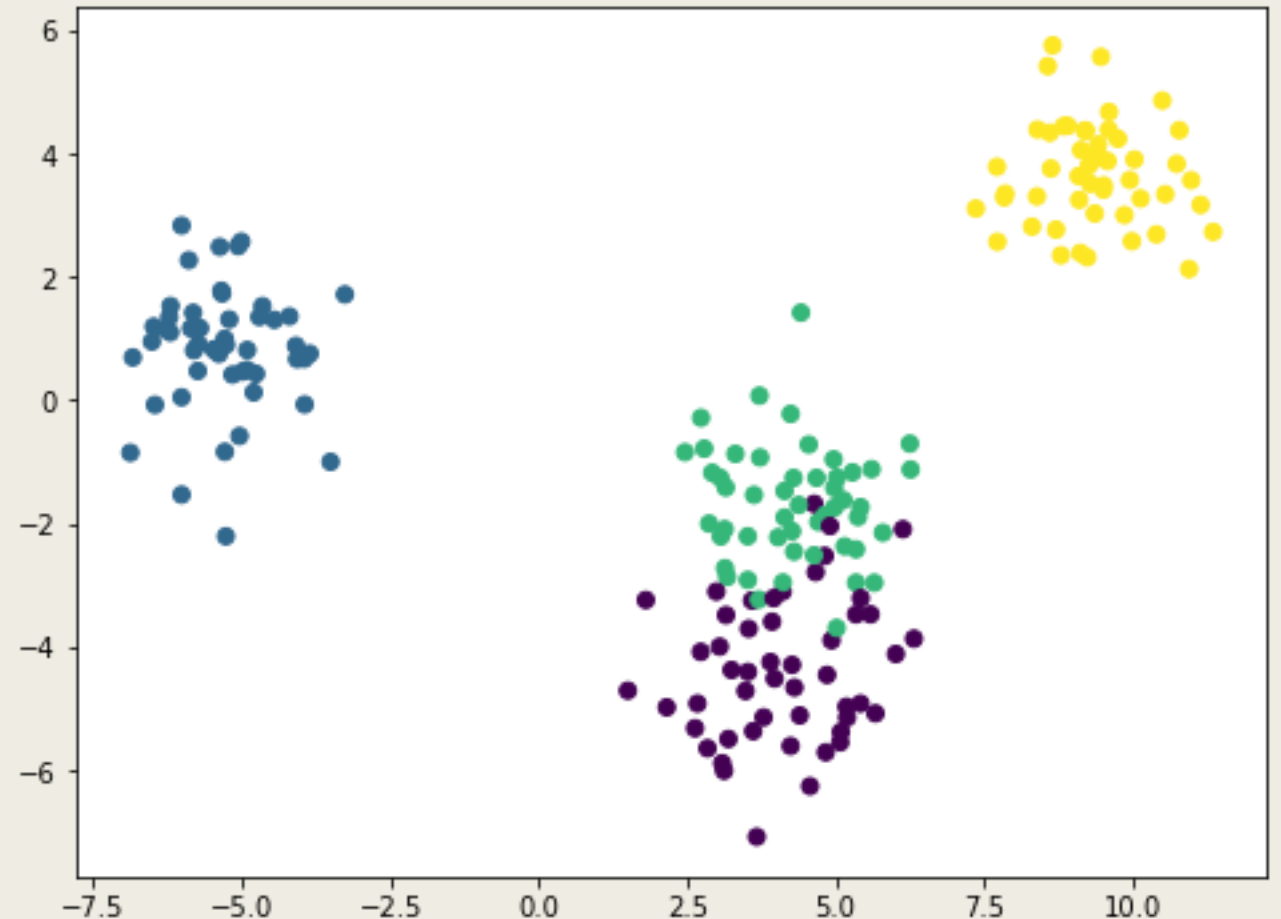
Clustering

- From the Elbow plot drawn between Within cluster sum-of-squares and number of centroids using the standardize data using K-Means, we can see the pre-defined clusters.
- we can find an optimal k value by finding the “elbow” of the plot, which is at 4.



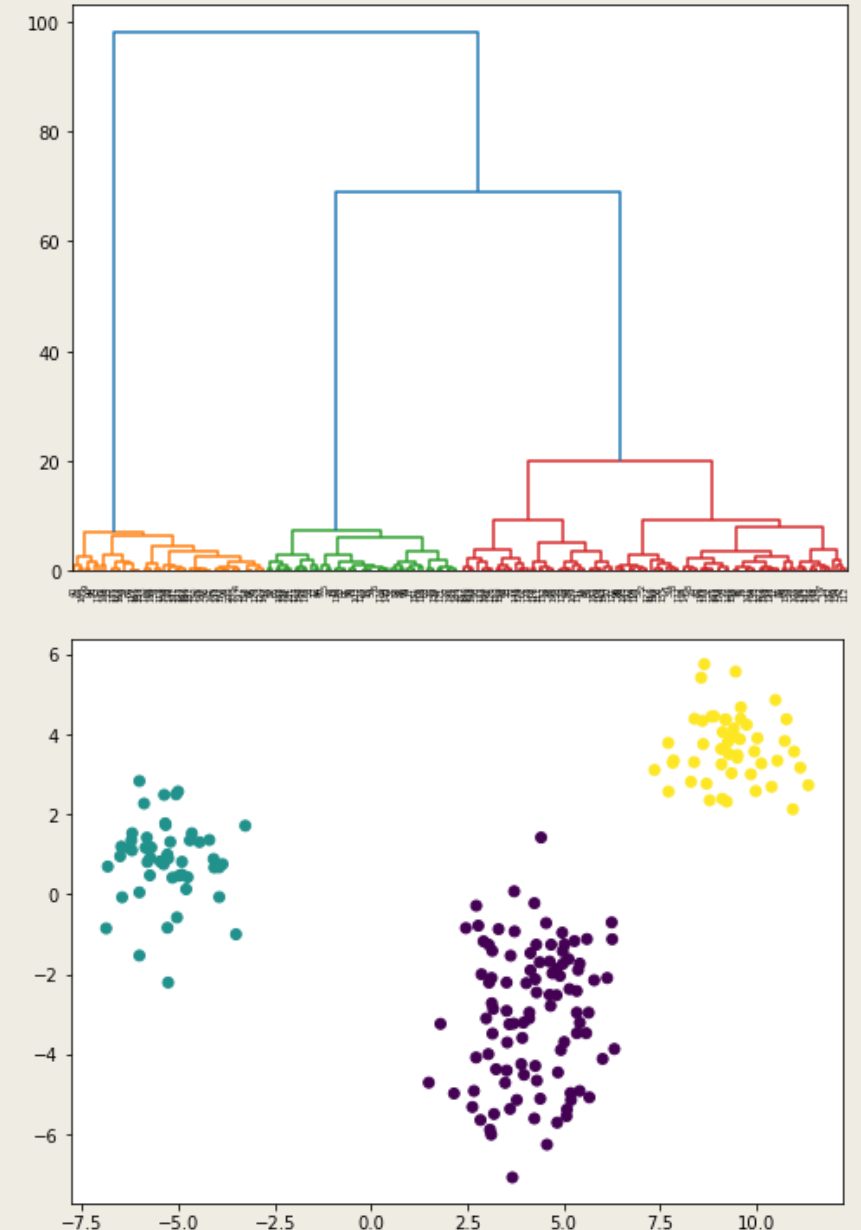
Clustering

- Here we got four clusters by the k value we found using the Elbow plot.
- These clusters are found for the sample of 200 with 4 centers.



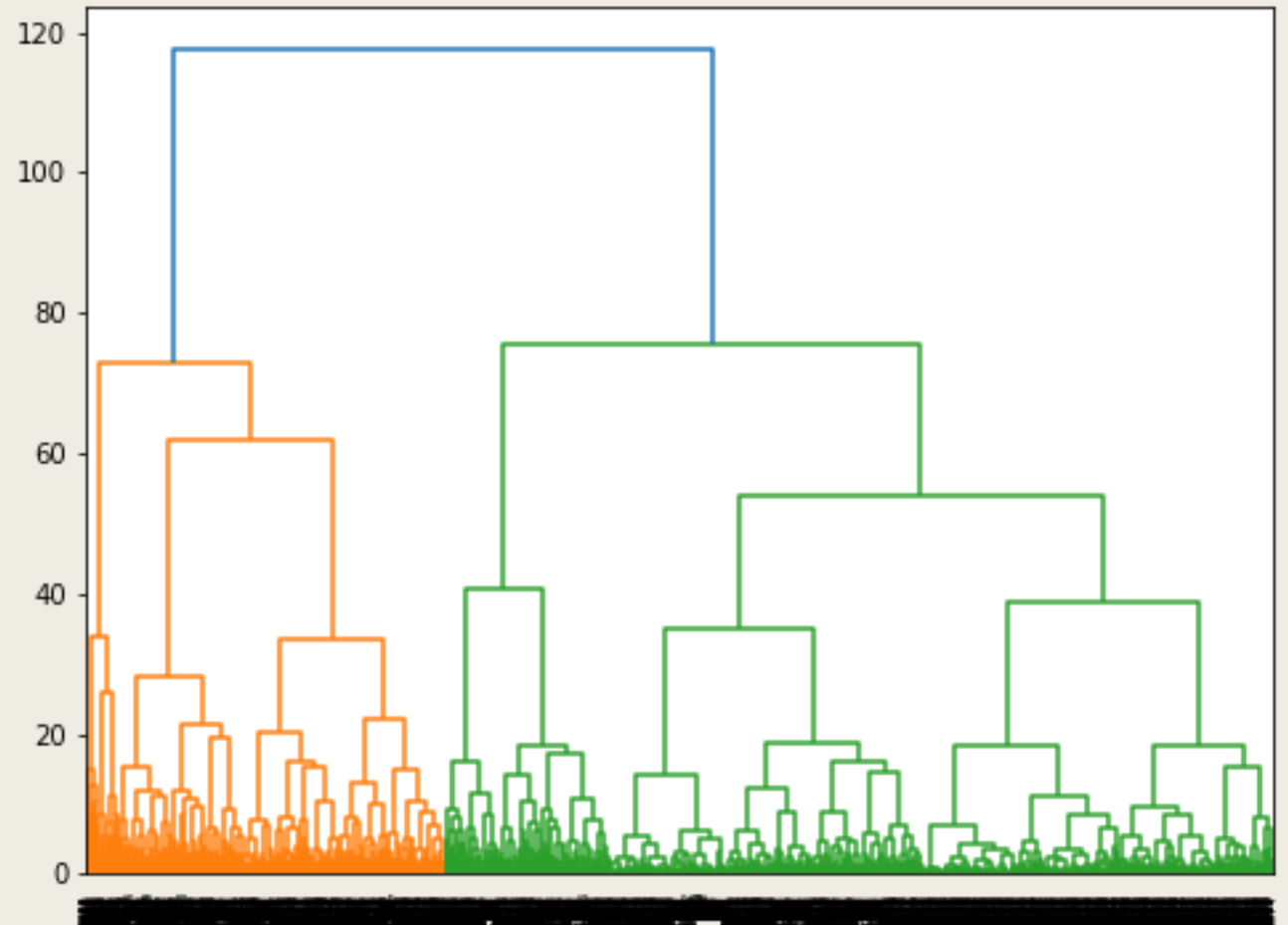
Clustering

- Using Dendograms of hierarchical clustering, we merged 3rd and 4th cluster together.
- By drawing the perpendicular lines through the blue vertical lines joining the 4 clusters together, we found the 3 clusters that we should consider.
- The scatter plot shows the newly found 3 clusters.



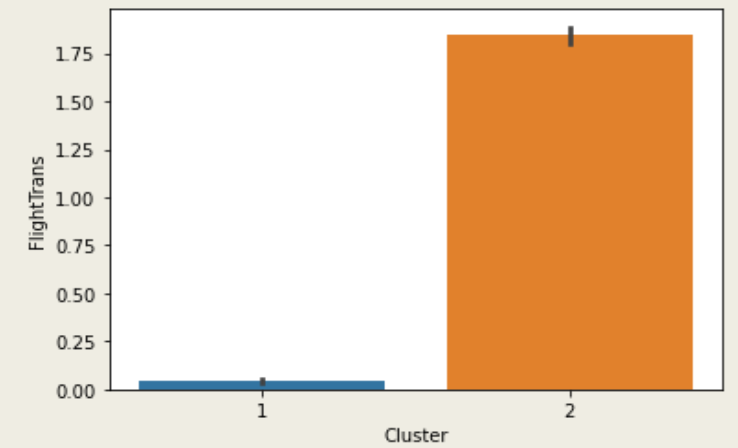
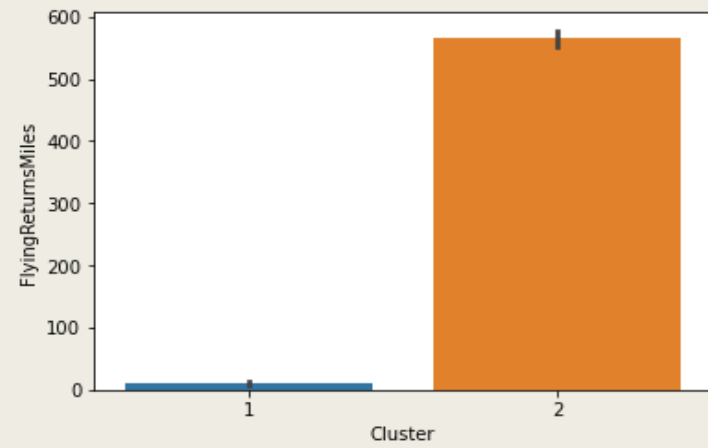
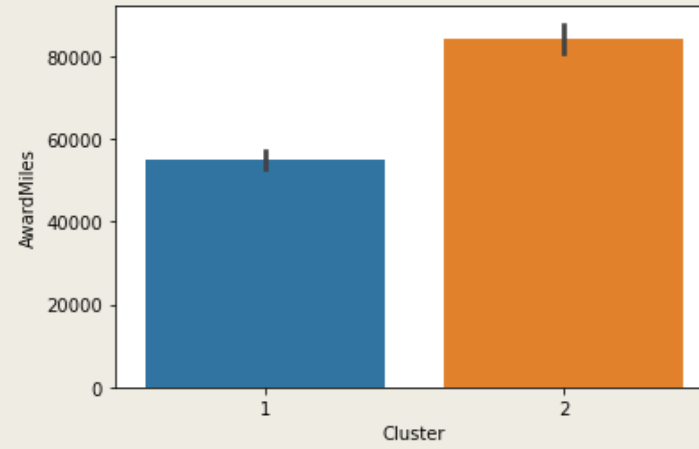
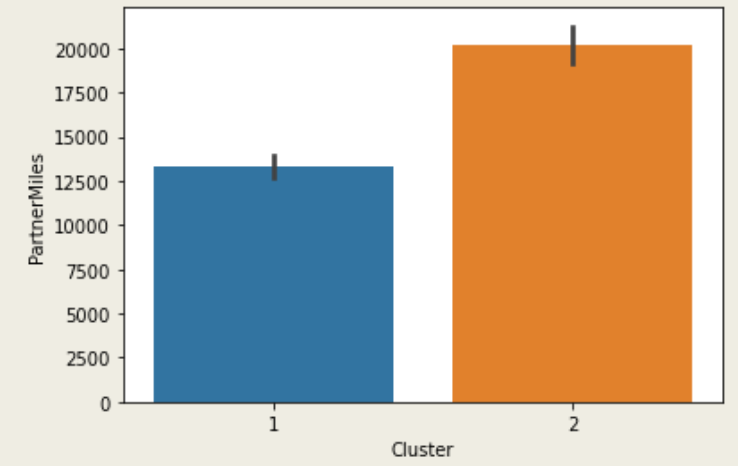
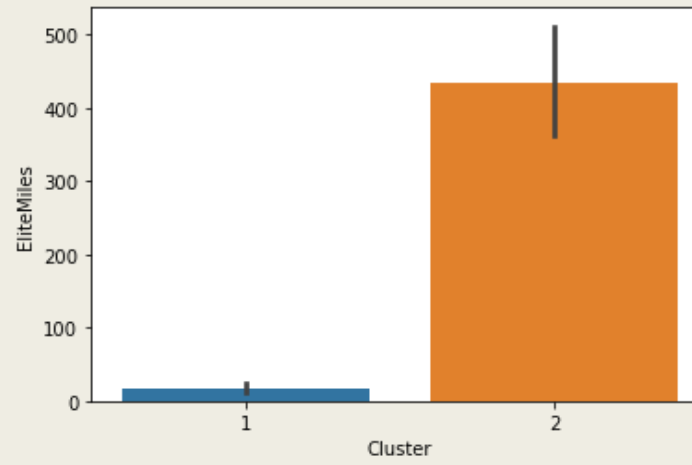
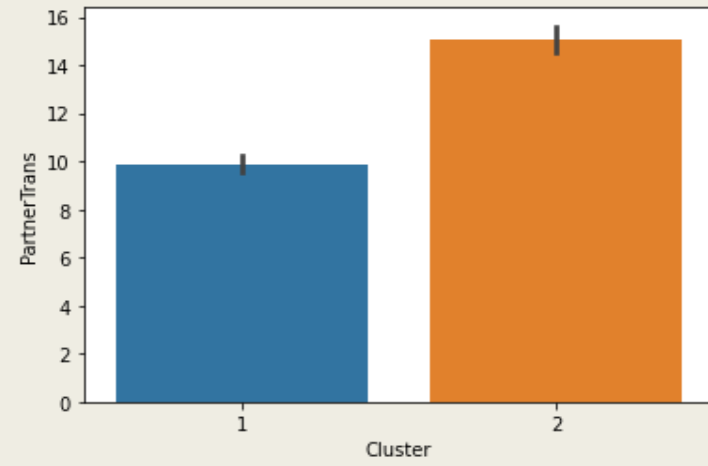
Clustering

By repeating the process to find cluster for the entire dataset, we found that the clusters are even reduced to 2.



Inference for clusters

- People in the cluster 2 has the highest amount of miles travelled which gives them a priority to attain Elite status.
- People in the cluster 1 has minimum number of balance miles to be eligible for Award Travel.
- More or less people of both the cluster stayed in the program above 4000 days.
- People in the cluster 2 earned large amount of miles by transacting using Partner's Service.
- Maximum number of people from cluster 2 travelled back using the same company's flight
- People in cluster 2 made the highest number of flight transaction.
- People from cluster 2 made the highest number of transaction using Partner's Service.



A person wearing a red coat is standing next to a vintage suitcase. The person's hand is visible, resting on the suitcase. The suitcase is light-colored with dark brown leather straps and buckles. The background is blurred, suggesting an outdoor setting. The text "THANK YOU" is overlaid in the center in a white, sans-serif font. The entire image is framed by a dark grey background with white L-shaped corner brackets in the top-left and bottom-right corners.

THANK YOU